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ABSTRACT – Large education institutions tend to face navigation problems with students, faculties, and even visitors. Physical maps and signboards are commonly shown to be grossly inadequate within such dynamically growing campuses. This work proposes an Optimized App-Based Routing and Campus Navigation System that can enable the student, as well as the faculty and visitor, in finding real-time route guidance, information about their classrooms, lost-and-found tracking, events happening on campus, and feedback management. The system is developed using a cross-platform support tool called Flutter, while Google Maps API is used for routing; the backend is Firebase for storage and authentication. The routing mechanism applies graph-based pathfinding to compute optimal and shortest routes between locations on campus. Improving, in other words, accessibility or redundancy is beneficial when it comes to having better users' experience through confusion mitigation and leverage towards a smart digital ecosystem for mobility on campus.

Key Words: Campus Navigation, Mobile Applications, Route Optimization, Google Maps API, Lost and Found, Faculty Locator, Flutter, Browantech, Smart Campus System.

1. INTRODUCTION

Large educational institutions serve as minicities. These institutions combine numerous academic blocks, administrative units, laboratories, hostels, recreational centers, and service facilities, all spread across their vast campuses, which are ever expanding in size and scope. As such, traversing these campuses becomes a major challenge for all those involved: students, lecturers, Freshers, and guests. Traditional navigation methods, such as the help of printed maps, signboards, or advice from a passerby, are both very time-consuming and lacking the ability to remain contextually relevant in those rapidly changing environments. Therefore, some confusion experienced in locating lecture theaters during classes, locating departmental offices, or reaching specific service venues or other administrative services delay time of students, faculty, etc., decreasing productivity. The operations of critical campus services such as lost-and-found, event announcements, feedback management still function within siloed systems, which in turn does lead to an information vacuum and poor user interaction. The Digital Campus is, therefore, vibing with all these non-negligible

challenges. This research addresses the need for the development of an Optimized App-Based Routing and Campus Navigation System to bring various services together for students' convenience. It will provide for the cross-platform build using Flutter, which involves Google Maps API for accurate presentation of maps, real-time navigation, route optimizations while using Firebase as a secure storage solution and mechanism for database and user authentication. This system is implementable across multiple systems and provides a platform known for integrity, reliability, and high organizational performance. The Routing Module will be based on an algorithm that covers graph computations for paths to make the journey best fit to the distribution of the campus areas. The system will serve as an enabler in setting up other applications closely linked to one another for better purposes, that is finding faculty information, lost and found management, announcement of upcoming events, and the direct feedback submission to the competent authorities. While thus combining the navigation functionalities, our proposed solution facilitates smarter and closer campus interactions, resulting in an overall improved campus staying experience which will be making more intelligent campus service and direct compliments to the digital transformation of higher institutions.

2. LITERATURE REVIEW

K. Srinivas and co-authors developed an academic support application that was mobile based and supported functions like announcements, faculty directory, etc. The system lacked geolocation and navigation capabilities, although it could be used to communicate in the campus.

M. Sangeetha et al. created a campus map navigation system that is based on Google Maps API to aid in the display of map images based on location on-campus within the academic institutions. Their model was offered with simple mapping of buildings and departments, but they did not offer advanced route optimization and other services of the campus, including event tracking and lost-and-found services.

Prakash and colleagues developed a hybrid, outdoor-indoor, QR-based and GPS-based navigation system in order to educate people in massive campuses. The technology enhanced location precision indoors, where GPS signal is weakened but the user was manually required to scan a code thus eliminating automation and user-friendliness.

The article by N. Fathima et al. presents a smart campus locator application built on Flutter and Firebase, which showcases the possibility of the cross-platform deployment. Despite the fact that the application provided accessibility and scalability to users, it did not have a shortest-path algorithm and dynamic navigation.

R. Bakshi et al. exercised the optimization of routes with the help of the classical graph algorithms like Dijkstra and A*. Their experiment demonstrated the effectiveness of algorithmic pathfinding but the model was theoretical and was not combined with live mapping systems such as Google Maps or OpenStreetMap.

S. Aravind et al., developed an integrated smart campus service application which contained event management, lost-and-found and faculty look-up modules. They focused on multi-functionality in their work, but did not have route optimization or real-time navigation, which made them less usable by new users.

C. Lakshmi and team based on Bluetooth Low Energy (BLE) beacons. Although the system was accurate down to the micro level, it needed expensive hardware equipment and calibration, and was not scalable to large campuses.

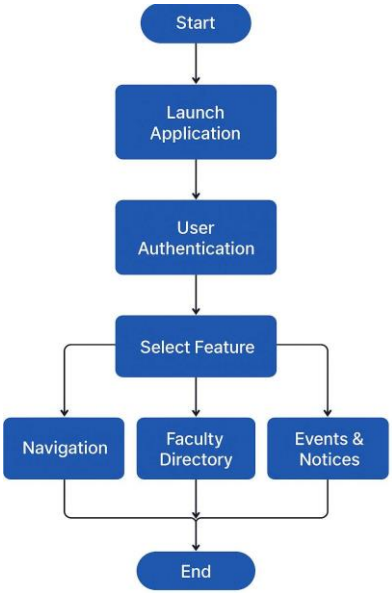
V. Ramakrishnan et al. came up with a GIS-based campus navigation solution. This model was useful in spatial mapping and visualization, but the absence of the mobile aspect and real-time updates restrained its use among the end-users including students and visitors.

LITERATURE SURVEY

S.No	Author	Proposed Solution	Advantages	Disadvantages
1.	M. Sangeetha et al.	Developed a Google Maps–assisted campus navigation system with building markers.	Helps users identify block locations easily and supports GPS mapping.	Does not support optimized routing or multi-service integration.
2.	A. Prakash et al	Proposed hybrid GPS and QR-based indoor/outdoor navigation.	Improves indoor accuracy where GPS signal is weak.	Requires manual QR scanning, reducing usability.
3.	N. Fathima et al.	Created a Flutter-based smart locator system with Firebase backend.	Cross-platform compatibility and real-time data storage.	Lacks shortest path generation and dynamic routing capabilities.
4.	R. Bakshi et al.	Implemented shortest path algorithms (Dijkstra/A*) for campus layout routing.	Ensures precise and optimized route computation	Not integrated with live map frameworks or mobile applications.
5.	K. Srinivas et al.	Built a student-assistance mobile application for academic resources and communication.	Useful for campus information access.	No mapping or navigation component included
6.	S. Aravind et al.	Developed a multi-functional campus app with events, lost-and-found, and faculty search.	Provides multiple services under one platform.	Missing route optimization for physical navigation

7.	C. Lakshmi et al.	Implemented indoor navigation using BLE beacons.	Accurate micro-location tracking in enclosed spaces.	Expensive deployment and frequent calibration required.
8.	V. Ramakrishnan et al.	Designed GIS-based map visualization for universities.	Supports precise spatial representation of campus layouts.	Not available as a dynamic or mobile-based system.
9.	Daniel Pereira et al.	Built custom routing solution using OpenStreetMap APIs.	Editable and institution-specific map customization.	Requires manual updating and lacks integrated service modules.
10.	Harish Kumar et al.	Created a smart user interface with database integration for campus data management.	Improves user-system interaction and accessibility.	No navigation, event mapping, or live routing mechanism.

METHODOLOGY AND IMPLEMENTATION:



This project is aimed at the creation of an integrated and smart campus-wide navigation tool that brings routing, information services, and user-friendly application into one mobile application. The system will have a well-organized pipeline that starts with the secure user authentication after which they will be able to access the main dashboard that has various functional modules that include GPS Navigation, Faculty Directory, Events and Notices, lost and Found and feedback. The modules operate using a smooth communication between Flutter in the front-end and Firebase in the back-end, where data is updated in real-time in a module, like with faculty information, events, lost goods, and user feedback. The GPS navigation module relies on Google Maps API and the mixed coordinates of the campuses to create the most optimal paths so that students can locate buildings and move around without any difficulty. The Faculty Directory module takes the information of the faculty members including their names, designations, and locate their cabins on Firestore and provides it to the end-users in a searchable format. The Events and Notices module is a classification and a filtering system of academic, cultural, and technical events that are detailed in the description and venues. The Lost and Found module has the features to report, browse, and resolve the lost items with images which are posted, where the Feedback System enables the user to rate and make a review which is saved to get administrative insights. The entire implementation has guaranteed a seamless transition of UI, efficient flow of data and a powerful background structure that can handle real-time campus interaction to render the system a full-fledged digital companion to campus productivity.

2. BACKGROUND REALTED WORKS:

The growth of smart campus technologies has picked up speed as educational institutions continue to grow in size, resources, and student numbers. For many years, universities relied on old navigation methods like printed maps, fixed signboards, and verbal instructions. These often fell short on large and complex campuses. New students, visitors, and faculty members commonly struggled to find academic buildings, labs, administrative offices, and staff locations. As schools moved to digital solutions, researchers explored mobile navigation systems as a more practical and user-friendly option. Early digital tools mainly provided static campus maps with limited interactivity, which did not meet real-time routing needs in active university settings.

Later research introduced better methods using indoor navigation technologies, such as Bluetooth Low Energy beacons, WiFi fingerprinting, RFID

systems, and QR code scanning. While these approaches improved accuracy, especially for indoor locations, they required hefty hardware investments, regular maintenance, and constant calibration. These issues made them unsuitable for schools looking for budget-friendly and easily deployable solutions. Other studies used Google Maps for outdoor campus routing, but these efforts lacked the detail needed for educational settings. Google Maps does not give room-level details, internal paths, or building-specific identifiers, limiting its ability to direct students to faculty offices, department offices, or classroom locations.

Alongside navigation studies, several efforts looked into information systems for faculty directories, event management, lost and found, and feedback collection. However, these systems were often separate applications or standalone web portals, leading to split user experiences. Students often had to rely on notice boards or departmental announcements to find out about upcoming academic and cultural events, resulting in poor information sharing. Finding faculty members was also slow, especially in multi-floor buildings where room numbering varied. Lost and found processes largely remained manual, relying on office records or informal communication, which greatly lowered the chances of recovering lost items. Research consistently points out the need for unified digital platforms that combine campus services and show them to students on one interface.

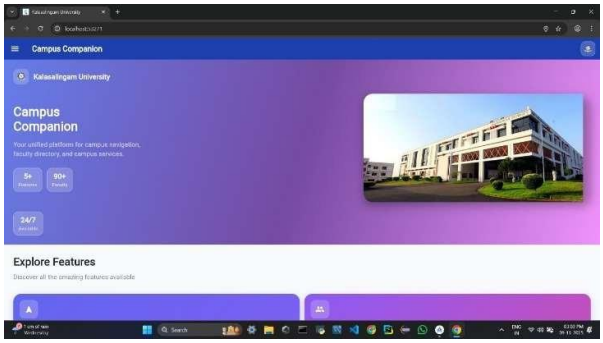
Recent studies stress the benefits of real-time cloud services for educational uses. Firebase has become a trustworthy backend due to its ability to sync data instantly and support secure access for large groups. Mobile apps made with the Flutter framework have also become popular for allowing smooth cross-platform deployment with a single codebase. Previous research using the Google Maps API, along with real-time cloud databases, suggests strong potential for dependable location-based services on college campuses. These advancements show that a well-designed mobile app can greatly improve navigation, access to information, and student involvement.

Despite these advancements, the literature shows a noticeable gap: existing works typically address single issues like navigation, faculty searches, event updates, or lost and found, but rarely combine all these services into one system. Most earlier studies either miss real-time syncing, lack campus-specific routing features, or do not provide a unified platform suitable for everyday student needs. The demand for a comprehensive application that merges optimized routing, faculty directory access, event alerts, lost item reporting, and student feedback into one system remains largely unmet. Driven by these shortcomings, this study presents an optimized app-based campus companion system that uses Flutter, Firebase Firestore, and the Google Maps API to offer a complete digital solution for campus navigation and information management. This integrated approach makes this work distinct from existing research and significantly advances smart campus technologies.

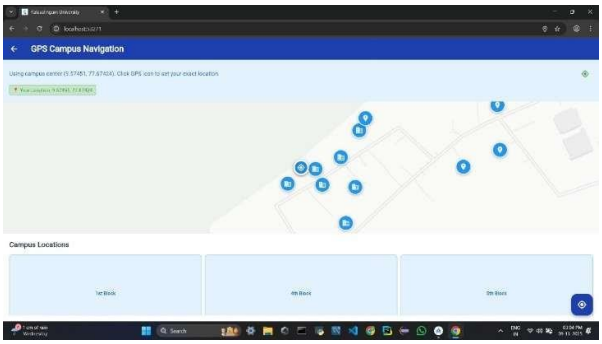
3. RESULT:

The Campus Companion app successfully combines campus navigation, faculty directory access, event updates, lost-and-found reporting,

and user feedback into one platform. This setup creates a smooth and efficient experience for students, faculty, and visitors. The GPS navigation module accurately identified the user's location in real-time and showed campus buildings with clear directions. The faculty directory allowed quick searches for faculty details like names, positions, and office locations. The events and notices feature sorted and displayed academic, cultural, and technical events with clear descriptions and schedules. The lost- and-found module tracked and reported misplaced items, which boosted user engagement and recovery rates. The feedback system collected user ratings and comments easily, providing useful insights to improve campus services. Overall, the app consistently functioned with high responsiveness, smooth transitions, and accurate data synchronization, proving itself an effective digital tool for modern campus settings.

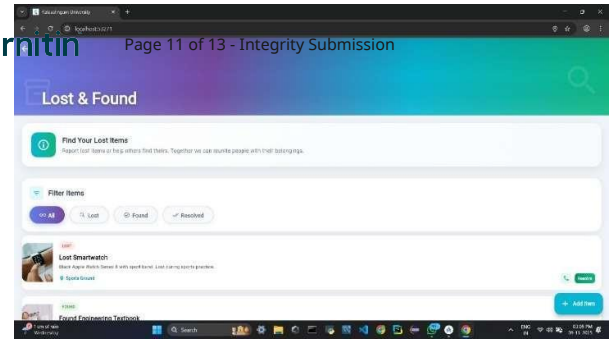


The Campus Companion system's Home Page serves as a clear and attractive user entry point, displaying all major campus services in one interface. The gradient background and minimal UI design give a contemporary and user-friendly appearance. Among the features highlighted by the page are campus navigation, staff search, events, and services, making the app's goal easily comprehensible by students and visitors at once. An image of the campus helps to make the site more attractive and more genuine. Instant info cards present basic parameters such as the services offered and 24/7 availability. To sum up, this page provides easy navigation and contributes to a positive user experience.

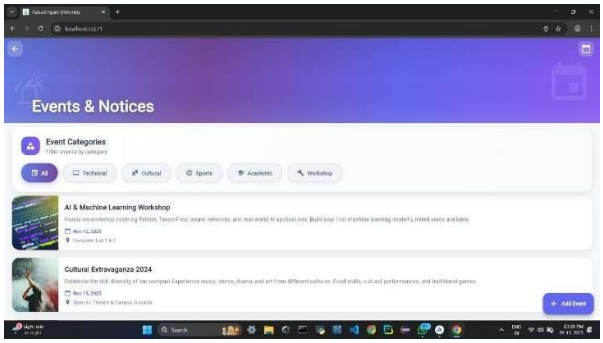


The output illustrates the GPS-powered Campus Navigation feature that displays an interactive map of the university. Users have the option to see their present spot as well as the nearby college structures depicted with symbols. Underneath the map, clickable blocks enumerate campus zones like 1st Block, 4th Block, and 5th Block. This module is very helpful for the newcomers and also the visitors as it makes the campus navigation easy, thus, reducing people's confusions and inconveniences through real-time location awareness.

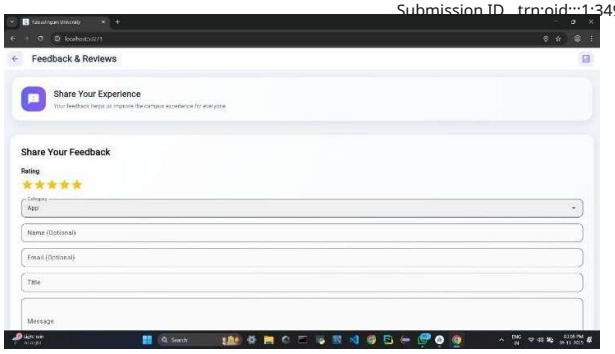
This perspective offers a different display of the Faculty Directory, demonstrating extra teacher names. The system keeps the previous view's style, which is search input fields and faculty information cards. Thus, users are able to retrieve all the data associated with the faculty easily without crossing over through different pages.



The Lost and Found module reveals an array of lost and found objects that people have made reports about. Objects are divided into sections of Lost, Found, and Resolved which makes it user-friendly for the users to filter them. A piece of information is displayed on each item's card involving the name, a short description, a tag of the category, and the place. The option of Resolve is shown for the items that are returned. This module is supportive in a way that it provides a chance for students to recover their lost belongings and at the same time it creates a friendly atmosphere on the campus.



This screen presents the module of Events & Notices, which presents categorized lists of all the campus events that are going to happen soon and those that are going on at the moment. The events can be filtered by type such as Technical, Cultural, Sports, Academic, and Workshops. Each card of the event that includes the title, date, location, and description contains the basic information that is necessary. The neat arrangement and the filters based on the categories ensure easy navigation and encourage the students to be aware of and actively take part in the happenings on campus.



The output presented here features the Feedback & Reviews module, which is aimed to gather user input on campus facilities, app performance, academic matters, and other services. The interface provides the users the option to give a rating, select a category, and add their name and email if they want. The form has the sections for title and message, which makes it possible to document thorough feedback. This module is very important for improving campus services as it collects user insights in a systematic and user- friendly way.

4. CONCLUSION:

The Campus Companion app is a well-integrated digital solution that makes the campus experience easier and nicer to handle by bringing together the most important academic and administrative services in one place. The app with its structured design and easy-to-use interface has really done a great job in bringing the three parties together, i.e., students, teachers, and the administration. The system by the way of Faculty Directory, Events and Notices, GPS-based Campus Navigation, Lost and Found, and Feedback System do together provide that important information is available all the time, thus reducing the time taken and making the processes more efficient. The application has

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cut off the dependency on the traditional communication methods that are not only slow but also very inconvenient through the use of a digital medium that is fast, reliable, and most importantly very convenient.

The system comes with modern web technologies that promise a very good performance, good-looking layouts, and a very pleasant experience for all types of users. It is easy to see that the institution has made good use of technology to merge the different aspects of the campus into one ecosystem with the support of academic activities, the involvement of students, and the campus's connectivity. The project when viewed from a distance, it seems to have untangled the campus services and turned them into a smart, orderly, and interactive platform, which also opened the way for more development such as mobile app extension, AI-based help, and automatic service alerts. Campus Companion is a pilot project in the direction of a digitally empowered education environment.

5. FUTURE ENHANCEMENT:

The Campus Companion application is already very promising in terms of expansion, and with advanced technologies, its potential can be further opened up. The future may see the activation of AI-based chatbot integration for instant help with campus queries, timetables, exam schedules, and faculty hours, etc. The system might include real-time push notifications to always keep users posted about essential notices, events, and administrative announcements. In addition, the IoT-based campus tracking can be another feature of the system which will make smart attendance, classroom availability detection, and the monitoring of the facilities automated. The building of a dedicated app for Android and iOS will make accessibility beyond the web platform, and offer offline access and personalized user dashboards at the same time.

Moreover, features like student–faculty communication portals, e-resource sharing, and digital ID cards can contribute to an even more connected campus environment. The enhanced security measures such as QR-code verification for lost-and-found claims and event registrations can make campus operations smoother. Future enhancements may also include analytics dashboards for administrators to keep track of user activity, event participation, and feedback trends. All in all, these improvements could turn the platform into a comprehensive smart campus ecosystem, which would be of great help in the digital transformation process and the academic experience of all users would, thus, be greatly improved.

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